

5. Project Development Phase

5.1. Data Cleansing & Scaler Construction (`preprocessing.py`)

Data cleansing and normalization is isolated into a reusable python module.

```
# Code snippet from preprocessing.py
```

5.2. Model Benchmarking and Comparison (`train.py`)

To prevent overfitting on the majority class (Approved = 25007), we apply manual oversampling to class 0 (Rejected = 97 in training split) to balance the partitions.

Four algorithms are trained on the resampled training partition and validated against a stratified 20% test partition:

5.3. Web Service Component (`app.py`)

Exposes three Flask controllers:

```
* --
```